LIOUVILLE'S THEOREM:

If the state of an ensemble changes with time, the position of phase points in the phase space will change with time. The motion of these phase points is totally governed by the canonical equation.

$$\dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i}$$

Where $i = 1, 2, 3 \ldots\ldots f$ for a system having f degrees of freedom 'H' is the Hamiltonian of the system and

$$H = H(q_1, q_2 \ldots\ldots q_f, p_1, p_2 \ldots\ldots p_f)$$

Liouvilles' theorem gives informations about the rate of change of phase density (ρ) in the phase space. The theorem may be stated in two parts:

Statements:

(i) The rate of change of density of phase

points in the neighbourhood of a moving phase point in Γ- space is zero.
ie It states the principle of conservation of density in phase space
ie $\frac{d\rho}{dt} = 0$

(ii) Any arbitrary element of region in Γ-space does not change with time inspite of displacements and distortions
ie It states the principle of conservation of extension in phase space.

$$\text{ie} \quad \frac{d}{dt}(\delta\Gamma) = \frac{d}{dt}\left\{\prod_{i=1}^{f} \delta q_i\, \delta p_i\right\} = 0$$

Proof: Consider an element of region in the phase space of hypervolume $\delta\Gamma$ where

$$\delta\Gamma = \delta q_1 \cdot \delta q_2 \cdots \delta q_f \cdot \delta p_1 \cdot \delta p_2 \cdots \delta p_f \quad \text{—} (1)$$

Now,
The number of phase points (δN) change with time and is given by using eqn (1)

$$\delta N = \rho \cdot d\tau = \rho \cdot \delta q_1 \delta q_2 \cdots \delta q_f \cdot \delta p_1 \delta p_2 \cdots \delta p_f \quad \text{—} (2)$$

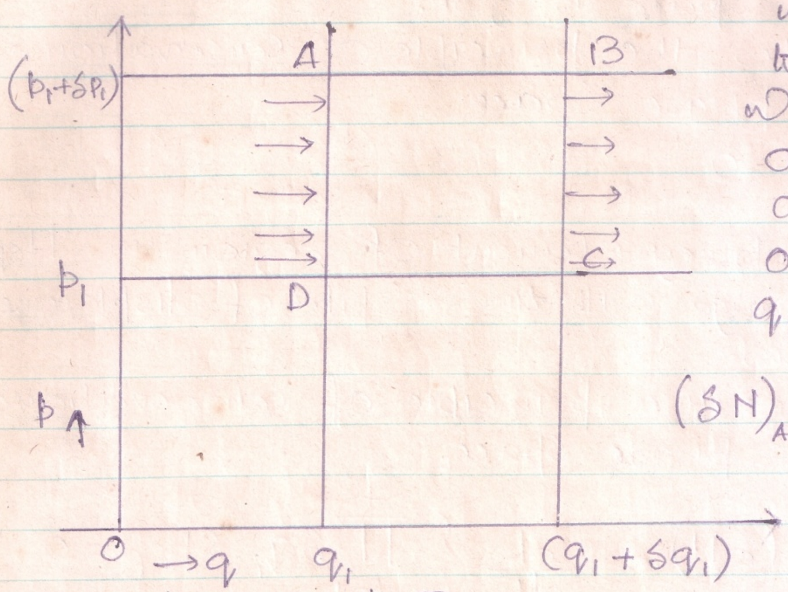
$$\therefore \frac{\partial}{\partial t}(\delta N) = \frac{\partial}{\partial t}(\rho \cdot \delta\tau) = \frac{\partial \rho}{\partial t} \cdot \delta\Gamma$$

Using eqn (1):

$$\text{ie} \quad \frac{\partial}{\partial t}(\delta N) = \frac{\partial \rho}{\partial t} \cdot \delta q_1 \delta q_2 \cdots \delta q_f \cdot \delta p_1 \delta p_2 \cdots \delta p_f \quad \text{—} (3)$$

The no. of phase points change because the no. of points entering the phase AD of the element of region is different than the no

Leaving the phase BC every sec.



with reference to the diagram which shows only two axes and considering only one co-ordinate q_1 we have, if

$(\delta N)_{AD}$ = No. of phase points entering AD per sec.

Then from eq^n (2):

$$(\delta N)_{AD} = (\rho \dot{q}_1 \cdot \delta q_2 \cdots \delta q_f \cdot \delta p_1 \cdots \delta p_f) \quad — (4)$$

Similarly if $(\delta N)_{BC}$ = No. of phase points leaving BC per sec.

We have from eq^n (4):

$$(\delta N)_{BC} = \left\{\left(\rho + \frac{\partial \rho}{\partial q_1} \cdot \delta q_1\right)\left(\dot{q}_1 + \frac{\partial \dot{q}_1}{\partial q_1}\delta q_1\right)\delta q_2 \cdots \delta q_f \cdot \delta p_1 \cdots \delta p_f\right\}$$

$$\therefore (\delta N)_{BC} = \left\{\rho \dot{q}_1 + \left(\rho \frac{\partial \dot{q}_1}{\partial q_1} + \dot{q}_1 \frac{\partial \rho}{\partial q_1}\right)\delta q_1\right\}\delta q_2 \cdots \delta q_f \quad — (5)$$

So that we may find the difference $(\delta N)_{AD} - (\delta N)_{BC}$ from eq^n (4) and (5) thus

$$\left\{(\delta N)_{AD} - (\delta N)_{BC}\right\} = -\left\{\left(\rho \frac{\partial \dot{q}_1}{\partial q_1} + \dot{q}_1 \frac{\partial \rho}{\partial q_1}\right)\delta q_1 \delta q_2 \cdots \delta q_f\right.$$

Similarly for the co-ordinate p_1 we have from eq^n (6):

$$\left\{(\delta N)_{AD} - (\delta N)_{BC}\right\} = -\left\{\left(\rho\frac{\partial \dot{p}_1}{\partial p_1} + \dot{p}_1\frac{\partial \rho}{\partial p_1}\right)\delta q_1 \delta q_2 \cdots \delta q_f \cdot \delta p_1 \cdot \delta p_2 \cdots \delta p_f\right\} \quad (7)$$

So that the total difference for the two coordinates is given by:

$$\text{Diff} = -\left[\left\{\left(\frac{\partial \dot{q}_1}{\partial q_1} + \frac{\partial \dot{p}_1}{\partial p_1}\right)\rho + \left(\dot{q}_1\frac{\partial \rho}{\partial q_1} + \dot{p}_1\frac{\partial \rho}{\partial p_1}\right)\right\}\delta q_1 \cdots \delta q_f \; \delta p_1 \cdots \delta p_f\right] \quad (8)$$

Now the above Difference is $\frac{\partial}{\partial t}(\delta N)$ so that we get from eq^n (3) and (8):

$$\frac{\partial \rho}{\partial t} = -\rho\left\{\left(\frac{\partial \dot{q}_1}{\partial q_1} + \frac{\partial \dot{p}_1}{\partial p_1}\right) + \left(\dot{q}_1\frac{\partial \rho}{\partial q_1} + \dot{p}_1\frac{\partial \rho}{\partial p_1}\right)\right\} \quad (9)$$

Equation (9) considers only two co-ordinates. Thus for all the co-ordinates eq^n (9) becomes

$$\frac{\partial \rho}{\partial t} = \left[-\sum_i \rho\left(\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i}\right) - \sum\left(\dot{q}_i\frac{\partial \rho}{\partial q_i} + \dot{p}_i\frac{\partial \rho}{\partial p_i}\right)\right] \quad (10)$$

Now the Canonical eq^{ns} are:

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \quad \& \quad \dot{p}_i = -\frac{\partial H}{\partial q_i}$$

H being the Hamiltonian. Then;

$$\frac{\partial \dot{q}_i}{\partial q_i} = \frac{\partial}{\partial q_i}\left(\frac{\partial H}{\partial p_i}\right) = \frac{\partial^2 H}{\partial q_i \partial p_i}$$

and

$$\frac{\partial \dot{p}_i}{\partial p_i} = -\frac{\partial}{\partial p_i}\left(\frac{\partial H}{\partial q_i}\right) = -\frac{\partial^2 H}{\partial p_i \partial q_i}$$

So that we have from the above relation

$$\frac{\partial \dot{q}_i}{\partial q_i} = -\frac{\partial \dot{p}_i}{\partial p_i}$$

i.e $$\frac{\partial \dot{q}_i}{\partial q_i} + \frac{\partial \dot{p}_i}{\partial p_i} = 0 \quad \text{—— (11)}$$

From eq^n (10) and (11) we obtain

$$\frac{\partial \rho}{\partial t} = -\sum_i \left(\dot{q}_i \frac{\partial \rho}{\partial q_i} + \dot{p}_i \frac{\partial \rho}{\partial p_i} \right)$$

Hence:

$$\left\{ \frac{\partial \rho}{\partial t} + \sum \dot{q}_i \frac{\partial \rho}{\partial q_i} + \sum \dot{p}_i \frac{\partial \rho}{\partial p_i} \right\} = 0 \quad \text{—— (12)}$$

Eq^n (12) is known as Liouville's Theorem.

The above eq^n may be written as below:

$$\frac{d\rho}{dt} = 0$$

Proof: We have $\rho = \rho(q_i, p_i, t)$

Then the total differential is given by

$$\frac{d\rho}{1} = \frac{\partial \rho}{\partial q_i} dq_i + \frac{\partial \rho}{\partial p_i} dp_i + \frac{\partial \rho}{\partial t} dt$$

$$\therefore \frac{d\rho}{dt} = \frac{\partial \rho}{\partial q_i}\frac{dq_i}{dt} + \frac{\partial \rho}{\partial p_i}\frac{dp_i}{dt} + \frac{\partial \rho}{\partial t}$$

i.e $$\frac{d\rho}{dt} = \dot{q}_i \frac{\partial \rho}{\partial q_i} + \dot{p}_i \frac{\partial \rho}{\partial p_i} + \frac{\partial \rho}{\partial t}$$

Taking into account all the co-ordinates the above eq^n becomes

$$\frac{d\rho}{dt} = \left\{ \frac{\partial \rho}{\partial t} + \sum_i \dot{q}_i \frac{\partial \rho}{\partial q_i} + \sum_i \dot{p}_i \frac{\partial \rho}{\partial p_i} \right\} \quad \text{—— (13)}$$

from eq^n (12) and (13) we obtain:

$$\boxed{\frac{d\rho}{dt} = 0}$$

(II) Consider a very small region of hyper volume $d\Gamma$ in the Γ space so that it may be assumed that the density (ρ) of the phase points in $\delta\Gamma$ is uniform throughout.
Let the no. of phase points in the hyper volume be δN

$$\therefore \quad \delta N = \rho \, d\Gamma$$

Differentiating w.r.to time we have

$$\frac{d}{dt}(\delta N) = \frac{d}{dt}(\rho d\Gamma) = \left\{\frac{d\rho}{dt} \cdot d\Gamma + \rho \frac{d}{dt}(\delta\Gamma)\right\} \quad \text{— (14)}$$

Now,

each phase point represents a definite System and as new system can neither be created nor destroyed.
We have:

$$\delta N = \text{constant}$$

$$\frac{d}{dt}(\delta N) = 0 \quad \text{— (15)}$$

Also we have; $\frac{d\rho}{dt} = 0$ — (16)

Substituting from eq^{ns} (15) and (16) in (14) we get

$$\rho \frac{d}{dt}(\delta\Gamma) = 0$$

Hence

$$\boxed{\frac{d}{dt}(\delta\Gamma) = 0}$$

Consequences of LIOUVELL'S THEOREM:

According to the principle of conservation of density in the phase space (a consequence of Liouvell's theorem) the density of a group of phase points remains constant along their trajectories in the phase space. If at any time the phase points are distributed uniformly in the phase space the phase points move in such a way that these will have uniform density of phase points at all times. Thus there is no crowding together of phase points in any particular region of phase space.

further,

According to the principle of conservation of extension in the phase space any arbitrary element of volume in the phase space bounded by a moving surface and containing a definite no. of phase points does not change with time despite the displacements and distorsions. The property of no ~~crowding~~ crowding of phase points in any particular regions of phase space and the constancy of the volume element in phase space with time indicates the truth of the postulates that

"The probability of finding a phase point in any particular region of phase space is directly proportional to the volume of that region."